

L5 SOD PYTHON PROGRAMMING GROUP WORK

1. Short Answer Questions

1. Define **Python programming language**.
2. Who created Python and in which year was it first released?
3. What is meant by **high-level programming language**?
4. Explain the meaning of **interpreted language**.
5. List **four uses of Python**.
6. Mention **three benefits of Python**.
7. What is **dynamic typing** in Python?
8. Explain the difference between **Python syntax and C/Java syntax**.
9. What is **indentation in Python** and why is it important?
10. Define **Python IDE**.
11. Give **three examples of Python IDEs**.
12. What is a **Python framework**?
13. Mention **two web development frameworks in Python**.
14. What is the role of **NumPy** in Python?
15. What is **automation in Python**?
16. Define a **data type** in Python.
17. List **five built-in data types** in Python.
18. What is a **variable**?
19. Write **two rules for naming Python variables**.
20. What are **comments in Python**?

2. Multiple Choice Questions (MCQ)

1. Python was created by:
A. James Gosling
B. Guido van Rossum

- C. Dennis Ritchie
- D. Bill Gates

2. Python is known as:
 - A. Low-level language
 - B. Machine language
 - C. High-level language
 - D. Assembly language
3. Which symbol is used for **single-line comments** in Python?
 - A. //
 - B. #
 - C. /* */
 - D. --
4. Which data type stores **text**?
 - A. int
 - B. float
 - C. str
 - D. bool
5. Which Python library is used for **data analysis**?
 - A. Django
 - B. Pandas
 - C. Flask
 - D. CherryPy
6. Which IDE is **beginner-friendly** for Python?
 - A. PyCharm
 - B. Thonny
 - C. Spyder
 - D. VS Code
7. Which operator is used for **exponentiation**?
 - A. ^
 - B. **
 - C. %
 - D. //
8. Which keyword is used to **define a function**?
 - A. func
 - B. define
 - C. def
 - D. function

9. Which framework is used for **web development**?

- A. Django
- B. NumPy
- C. Pandas
- D. Matplotlib

10. Which data structure stores **key-value pairs**?

- A. List
- B. Tuple
- C. Dictionary
- D. Set

3. True or False Questions

- 1. Python code must always be compiled before running.
- 2. Python uses indentation to define blocks of code.
- 3. Lists in Python are immutable.
- 4. Dictionaries store data as key-value pairs.
- 5. Lambda functions have a name.
- 6. PyCharm is a Python IDE.
- 7. Sets allow duplicate values.
- 8. Python supports object-oriented programming.
- 9. Functions improve code reusability.
- 10. Generators use the **yield** keyword.

4. Practical / Programming Questions

- 1. Write a Python program to **print "Hello World"**.
- 2. Write a Python program that **asks the user for their name and age**, then prints a greeting message.
- 3. Write a Python program that **adds two numbers**.
- 4. Write a Python program to check whether a number is **even or odd** using an if...else statement.

5. Write a Python program that **calculates the square of a number using a function**.
6. Write a Python program that **creates a list of fruits and prints each fruit**.
7. Write a Python program that **uses a lambda function to double a number**.
8. Write a Python program that **filters even numbers from a list using filter()**.
9. Write a Python program that **calculates factorial using recursion**.
10. Write a Python program that **uses a loop to print numbers from 1 to 10**.

5. Essay / Long Questions

1. Explain the **benefits of Python programming language**.
2. Describe the **criteria for selecting Python tools**.
3. Explain the **different Python IDEs** and their features.
4. Discuss the **applications of Python in data science and automation**.
5. Explain the **different types of functions in Python** with examples.